# Datasets

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The datasets used in the course exercises are organized by week. Click on the filenames to download the data files.

## 1 Week 45

| Dataset | Exercise | Description |
|---|---|---|
| averag-prope-amino-acids.xlsx | Average properties of amino acids and proteins. | Amino acid frequency data |
| uv-spec-apo-holo-myo.csv | UV spectrum of apo- and holo-myoglobin | UV spectra for apo and holo myoglobin |
| trypt-absor-fluor-emission.xlsx | Tryptophan absorption and fluorescence | Tryptophan fluorescence emission data |
| trypt-absor-fluor-extinction.xlsx | Tryptophan absorption and fluorescence | Tryptophan fluorescence extinction data |
| trypt-absor-fluor-ph2.xlsx | Tryptophan absorption and fluorescence | Tryptophan fluorescence at pH 2 |
| trypt-absor-fluor-ph7.xlsx | Tryptophan absorption and fluorescence | Tryptophan fluorescence at pH 7 |
| extin-coeff-human-myogl.txt | Extinction coefficient of human myoglobin | Titin protein sequence data |
| the-fluor-protei-mcherr.csv | The fluorescent protein “mCherry”. | mCherry fluorescent protein spectra |
| protei-blood-plasma.xlsx | Proteins in blood plasma. | Protein concentration in blood plasma data |

## 2 Week 46

| Dataset | Exercise | Description |
|---|---|---|
| dialys-exper.xlsx | Dialysis experiment | Dialysis experiment data |
| adp-bindin-pyruva-kinase.csv | ADP binding to pyruvate kinase. | ADP binding to pyruvate kinase data |

| Dataset | Exercise | Description |
|---|---|---|
| inter-bindin-data.xlsx | Interpretation of binding data. | Binding data for interpretation |
| deter-type-streng-coope.xlsx | Determination of type and strength of cooperativity | Cooperativity determination data |
| reaction-orders.xlsx | Determination of reaction orders. | Reaction order determination data |
| binding_data_sq.xlsx | A tricky protein binding. | Binding data for a tricky protein. |
| exp_decay_data.xlsx | — | Exponential decay data for regression exercises. |

## 3 Week 47

| Dataset | Exercise | Description |
|---|---|---|
| deter-reacti-order-activ.csv | Determination of reaction order and activation energy | Reaction order and activation energy data |
| design-enzyme-kineti-exper.xlsx | Design of an enzyme kinetics experiment | Enzyme kinetics experimental design data |
| analys-data-set-obeyin.xlsx | Analysis of a data set obeying Michaelis-Menten kinetics | Data set analysis for enzyme kinetics |
| kinetics_data.xlsx | Lab Exercise Analysis: Michaelis Menten | Use your own data if you have it, use this dataset if you don't or your data is of poor quality. |

## 4 Week 48

| Dataset | Exercise | Description |
|---|---|---|
| enzyme-inhib-i.xlsx | Enzyme inhibitors (I) | Enzyme inhibition data (part I) |
| enzyme-inhib-ii.xlsx | Enzyme inhibitors (II) | Enzyme inhibition data (part II) |
| enzyme-behav-atcase.csv | Enzymatic behaviour of the enzyme ATCase | Aspartate transcarbamoylase enzyme behavior data |

## 5 Week 49

| Dataset | Exercise | Description |
|---|---|---|
| deter_delta_h_data.xlsx | Determination of $\Delta H^\circ$ for a reaction from the van't Hoff plot. | Determination of $\Delta H^\circ$ for a reaction dataset (Excel format) |
| diff_q_keq_data.xlsx | Difference between $Q$ and $K_{eq}$ | Dataset of time and product concentrations. |

## 6 Additional Datasets

*These datasets are available but not yet assigned to specific weeks.*

| Dataset | Exercise | Description |
|---|---|---|
| chlorophyll_adsorption.xlsx | — | Chlorophyll absorption data (Excel format) |
| reverse_reaction.xlsx | — | Reversible reaction kinetics data (Excel format) |